

Forecasting Realized Volatility of Crude Oil Futures Prices based on Machine Learning^{*}

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Abstract

Extending the popular HAR model with additional information channels to forecast realized volatility of WTI futures prices, we show that machine learning generated forecasts provide better forecasting quality and that portfolios which are constructed with these forecasts outperform their competing models and resulting in economic gains. By analyzing the selection process, we show that the information channels vary across the forecasting horizon. Variable selection produces clusters and provides evidence that there are structural changes in relation to the importance of information channels.

Keywords: Forecasting, Crude oil, Realized volatility, Exogenous predictors, Machine learning

JEL classification: C22; C45; E37; Q47

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1. Introduction

As a tangible quantification of uncertainty, measuring, understanding, and modeling volatility is a cornerstone of comprehending commodity markets in their functionality, price discovery, and structure of connectedness to other markets. Commodity markets are inevitably linked to the economic activity and well-being of global and local economies, as they embody the backbone for supply and demand of primary materials and feedstock for production, biomaterial for agricultural and food industries, and energy commodities, among others. Energy commodity markets, in particular, play an important role in research, as their understanding is of utmost importance for pricing, security, and planning of supply, as well as policy design and implications. The link between oil prices and economic growth is well documented and discussed in the literature (Hamilton, 2003, Kilian, 2009, Charfeddine et al., 2020). In this study, we present novel evidence on the impact of exogenous information channels on the realized volatility of crude oil futures prices. Building on the popular Heterogeneous Autoregressive model (HAR) by Corsi (2009), we show that models that incorporate variables that map the volatility of the equity channel, sentiment, or foreign exchange market result in more accurate predictions of realized volatility than the baseline HAR. In addition, we find that machine learning variants of the HAR model outperform baseline HAR models by selecting the most useful exogenous predictors from the set of different information channels; both in view of forecasting performance and economic gain. We also analyze in detail the time variation and importance of variables in predicting the volatility of the oil price. When comparing the variable selection across forecasting horizons, we find a positive relation between the forecasting horizon and number of different information channels. Looking at the variable importance of those channels, we find particularly FX to add to the information contained in past realized volatility.

We contribute to the literature on the connectedness of crude oil markets. It is argued that there exists an information transmission between the volatility of markets that leads to a co-movement of the global oil market (Reboredo, 2011, Kaufmann & Banerjee, 2014, Luo et al., 2023). Furthermore, WTI is taking a leading role as the global benchmark,

while other energy markets such as Brent crude oil follow these movements (Elder et al., 2014, Klein, 2018). Adding to the complexity of this research, studies show that exogenous factors and information channels that cannot be directly associated with energy markets also affect the behavior of price and volatility (e.g. Nguyen & Walther, 2020, Dudda et al., 2022). In particular, other information channels, such as equity markets (Degiannakis & Filis, 2017, Haugom et al., 2014, Luo et al., 2020) or Google Search Volume (Afkhami et al., 2017) are shown to provide valuable information to forecast oil volatility with extended HAR models.

In contrast to previous studies, we test the impact of volatility of different equity markets, such as the Dow Jones Industrial, the CAC40, and the S&P 500 among others, commodity futures markets, e.g. Natural Gas and Gold, foreign exchange (FX) markets, sentiment and volatility indicators such as OVX and VIX, and finally Google trends regarding oil price related search terms on realized volatility of WTI crude oil. As we utilize an overall number of 28 exogenous channels from five categories, resulting in a total of 84 exogenous HAR factors, model parameter parsimony is forgone. We extend the baseline HAR model with each category of exogenous variables separately. Then, we introduce a “kitchen sink” approach to obtain the last HAR extension, in which we simultaneously include all exogenous variables. Therefore, the number of parameters in each HAR extension ranges from 16 to 31 for separated channels and to 88 for the model that incorporates all channels, since each channel element is included in daily, weekly, and monthly averages. These highly parameterized models are predestined to be combined with machine learning (ML) methods to filter for the most relevant contributing factors and to overcome multicollinearity within and across information channels. Additionally, we are tracking the selection of variables across different forecasting horizons for inference on possibly time-varying impact of equity and sentiment channels on the forecasts of the realized volatility of crude oil futures prices.

In addition, we contribute to the literature on modeling and forecasting the (realized) volatility of oil prices. It is well documented that crude oil volatility behavior is characterized by its time-varying nature, including structural breaks or other time-varying

effects (Fong & See, 2002, Fattouh, 2010, Nomikos & Pouliasis, 2011, Klein & Walther, 2016). This motivates the adaptation of models covering these properties. In contrast, some studies show that more sophisticated HAR models based on stylized volatility facts do not necessarily outperform the HAR version without extensions or exogenous factors (Sévi, 2014, Prokopczuk et al., 2016, Degiannakis et al., 2022). Related to our article, Ma et al. (2018) and Zhang et al. (2019a) show that ML improves volatility forecasts for crude oil prices using a large set of exogenous predictors. We extend these studies by comparing various popular ML-based variable selection methods with large sets of predictors for crude oil forecasts and their economic gains. These methods include the widely applied LASSO and Bayesian Model Averaging (BMA) as well as novel approaches to HAR models with Bootstrap Aggregation and Bagging (BAG) and Stochastic Search Variable Selection (SSVS). In this regard, we also contribute to the general literature on modeling and forecasting with HAR models.

The remainder of the paper is structured as follows. Section 2 introduces the HAR models, their extensions, and the variable or adaptive selection methodologies of the study. Section 3 describes the data sets and presents some preliminary analysis of exogenous information channels. Section 4 reports and discusses the results of the empirical study, while Section 5 concludes.

2. Model and Methodology

2.1. *Heterogeneous autoregressive volatility models with adaptable predictors*

In order to measure time-varying daily volatility based on the available high-frequency data sets, we calculate the realized volatility RV_t of day t as

$$RV_t = \sum_{i=1}^m r_{t,i}^2, \quad (1)$$

where m is the number of observed intra-day returns $r_{t,i}$ with $i = 1, \dots, m$. This i -th intra-day return of day t is calculated as logarithmic difference of intra-day prices as

$$r_{t,i} = 100 \times (\log P_{t,i} - \log P_{t,i-1}),$$

where $P_{t,i}$ and $P_{t,i-1}$ are two consecutive intra-day prices. We choose to sample prices at five minute intervals.¹

We follow the approach of Corsi (2009) and exploit the autoregressive character of daily realized volatility by cascading its memory structure in a short-, medium-, and long-term component that yields the Heterogeneous Autoregressive (HAR) model for realized volatility. Our baseline HAR model uses the specifications described in Corsi & Renò (2012). Here, we run a different regression model for each forecast horizon h . Therefore, long-term forecasts do not have to rely on the relative weights of the predictions for one day ahead for $h > 1$ (Ederington & Guan, 2010). In particular, we define

$$\log RV_{t+h}^{(h)} = \frac{1}{h} \sum_{j=1}^h \log RV_{t+h-j+1} \quad \text{and} \quad (2)$$

$$\log RV_t^{(h)} = \frac{1}{h} \sum_{j=1}^h \log RV_{t-j+1}, \quad (3)$$

where $h \in \{1, \dots, 22\}$ denotes the days-ahead forecasting horizons. We distinguish $\log RV_{t+h}^{(h)}$, which is the average realized volatility for time $t+1$ to $t+h$, from $\log RV_t^{(h)}$, which is the average realized volatility for time $t-h+1$ to t . Lastly, $\log RV_t$ is the realized volatility at time t and is equivalent to $\log RV_t^{(1)}$.

Our HAR baseline model reads as

$$\log RV_{t+h}^{(h)} = \beta_0 + \beta_1 \log RV_t + \beta_2 \log RV_t^{(5)} + \beta_3 \log RV_t^{(22)} + u_{t+h}^{(h)}, \quad (4)$$

where β_0 , β_1 , β_2 , and β_3 are real-valued coefficients corresponding to the intercept and the short-, medium-, and long-term autoregressive impact of realized volatility and u is an i.i.d. error term with zero mean. In the standard HAR defined in Eq. (4), RV_t then refers to the realized volatility of the previous day, $RV_t^{(5)}$ to the average realized volatility over the past five days, and $RV_t^{(22)}$ refers to the 22-day average.

¹This choice of sampling frequency is widely adapted in literature as it poses a trade-off between possible microstructure noise, where sampling frequency is too high, and information loss, where sampling frequency is too low (Aït-Sahalia & Yu, 2009, Liu et al., 2015).

As outlined in the Introduction, there exist numerous extensions to the HAR model, in particular in view of modeling realized volatility of commodities.² In this research, we focus on an extension of the HAR model framework with exogenous factors following the notion of Zhang et al. (2019b) and Luo et al. (2020).

We consider five classes of exogenous predictors that might impact the volatility behavior of crude oil futures prices. The first class includes equity market volatilities of eight major economies. The second class refers to market sentiment and uncertainty measures. The third and fourth class of exogenous variables comprise realized volatility of related commodity futures—with a documented economic link to crude oil futures—and foreign currency (FX) markets, respectively. The last class includes a sentiment index constructed based on the popularity of different Google search queries. Details on these exogenous predictors are described in detail in the data description in Section 3.

We extend the basic HAR of Eq. (4) with lagged daily, weekly, and monthly aggregates of exogenous variables $X_{k,t}$ as follows:

$$\begin{aligned} \log RV_{t+h}^{(h)} = & \beta_0 + \beta_1 \log RV_t + \beta_2 \log RV_t^{(5)} + \beta_3 \log RV_t^{(22)} \\ & + \sum_{k=1}^{N_C} \alpha_{1,k} X_{k,t}^C + \sum_{k=1}^{N_C} \alpha_{2,k} X_{k,t}^{C,(5)} + \sum_{k=1}^{N_C} \alpha_{3,k} X_{k,t}^{C,(22)} + u_{t+h}^{(h)}. \end{aligned} \quad (5)$$

where N_C refers to the number of exogenous predictors from category C and $\alpha_{1,k}$, $\alpha_{2,k}$, and $\alpha_{3,k}$ for $k = 1, \dots, N_1$ are the corresponding coefficients of the daily, weekly, and monthly averages or aggregates of the k -th component of exogenous predictor category C .

Using model Eq. (5) in combination with our predictor classes, we obtain five different models, which we label: HAR-Equity, HAR-Sentiment, HAR-Commodity, HAR-FX, and HAR-Google. For HAR-Equity, we use realized volatilities sampled at a matching 5-minute interval. We employ squared daily returns of commodity futures and FX rates, as well as the U.S. dollar index to proxy volatilities of commodity futures and FX markets.

²Examples can be found in Sévi (2014), Klein & Todorova (2021), Luo et al. (2022), and Degiannakis et al. (2022) among others.

The weekly and monthly volatilities of the commodity futures and FX markets are defined as $r_{i,t}^{2,(5)}$ and $r_{i,t}^{2,(22)}$ which are computed in the same way as the weekly and monthly realized volatilities.

In addition to the aforementioned extensions of the baseline HAR model using different asset classes and information channels, we employ a model containing *all* exogenous factors of this study, which we label HAR-All. This HAR-All contains 84 exogenous parameters, three endogenous parameters, and the intercept. As we use different ML approaches for exogenous variable selection, HAR-All poses an additional opportunity to compare these differing techniques in terms of forecasting performance and economic valuation. For further robustness of our findings, we also employ a model class using the information from the first principal component of each asset/information channel for each time horizon. This model class is labeled HAR-PCA.

In order to have a very parsimonious benchmark, we also include an autoregressive (AR) model with lag 1. Doing so allows us to assess to what extent additional and aggregated information adds predictive power over the information already present in the last observed realized volatility.

Finally, we also add an augmented AR(1) model, AR(1)-X, which includes additional information sets similar to the extended HAR in Eq. 5.

2.2. HAR models with variable selection approaches

There exist some popular approaches in the literature with regard to selecting predictors and shrinking variable dimensions. Our motivation for implementing a range of these methodologies is two-fold. Firstly, information processing is computationally costly, and the inclusion of insignificant predictors leads to an increasing need of information processing capacity. Investors and researchers alike prefer to focus on relevant information channels through variable selection schemes to simplify economic models (Gabaix, 2014, Luo & Young, 2016, Zhang et al., 2019b). Secondly, many exogenous and powerful predictors, such as macroeconomic and financial market indicators, are highly correlated, which can lead to an over-fitting problem and overshadowing of significant parameters through present multicollinearity. With variable selection methods, only the most rele-

vant variables are selected as predictors, thus reducing the risk of over-fitting issues and improving the forecast accuracy (Campbell & Slack, 2008, Korobilis, 2017, Korobilis & Koop, 2020). Analyzing the systematic selection of variables also allows us to identify breakpoints, offering additional insight on the variability of the statistical and economical benefit of including exogenous predictors in HAR frameworks.

As outlined in the introduction, we construct a set of competing HAR models based on several variable selection approaches, including the well-known LASSO, BMA, BAG, and SSVS. These approaches can be further categorized as Bayesian algorithms that include predictor selection with BMA and SSVS approaches and machine learning with LASSO and BAG falling into this category. We briefly introduce the selection approaches and use the abbreviation for the selection method as the prefix of the respective HAR variant in what follows.

For the LASSO-HAR, we employ the R-package ‘glmnet’ of Friedman et al. (2010) to compute and select the coefficients of the HAR models and its extensions introduced above. The coefficients are obtained by solving

$$\hat{\beta}^{LASSO} = \arg \min_{\beta_0, \gamma} \left(\frac{1}{2T} \sum_{t=1}^T (y_{i,t} - \beta_0 - Z_{t-1}\beta)^2 + \lambda \sum_{j=1}^{3K} |\beta_j| \right),$$

where Z_{t-1} is a K -dimensional column vector of all the predictor variables in the different HAR models above and β is a K -dimensional row-vector with the respective coefficients. The non-negative regularization or penalty parameter λ is selected given the minimum mean cross-validated error. Eventually, $\hat{\beta}^{LASSO}$ is the K -dimensional vector of the estimated coefficients from the LASSO regression.

The BMA tackles the problem of model uncertainty and model selection from a Bayesian perspective. Over a set of models $\mathcal{M}_i$ for $i = 1, \dots, 2^K$ (i.e. all combinations of K predictors), the BMA determines the posterior probability of each model. The approach then averages the models using their posterior probability. For the BMA-HAR model, we follow the setting of Fernández et al. (2001). We specify the g -prior with $g = 1/K^2$. Markov-chain Monte Carlo methods are used for the estimation of the BMA-HAR models according to Madigan et al. (1995) and Dellaportas et al. (2002).

The BAG approach reduces the variance of the forecasts by averaging the randomness of variable selection and has been widely applied in the areas of macroeconomic and financial forecasting (Lee & Yang, 2006, Inoue & Kilian, 2008, Huang & Lee, 2010, Audrino & Medeiros, 2011, Stock & Watson, 2012). With bagging, the bootstrap method is employed to generate a large amount of samples from the original data to reevaluate the selection of predictors and to produce the bootstrap replications of forecasts, and the forecast realized volatilities are obtained by averaging the forecasting values based on the bootstrap samples. In particular, we use the same setting and critical values as in Inoue & Kilian (2008) and Ribeiro (2017). The optimal predictor set is data-dependent in the sense that the pre-tests in the extended linear HAR model above. The bagging approach is as follows:

- Arrange the set of predictors Z_t , $t = 2, \dots, T$, in form of a matrix M of dimension $(T - 1) \times K$, where K is the number of predictors in Z_t .
- Create bootstrap samples of the form $\{Z_{(i)2}^*, \dots, Z_{(i)T}^*\}$, for $i = 1, \dots, B$, by drawing blocks of m rows of M with replacement, where the block size m is selected to capture the possible dependence in the error term of the realized volatility for oil futures. Based on Ribeiro (2017), we choose $B = 100$ replications and a block size of $m = \lfloor T^{1/4} \rfloor$ for the moving block bootstrap procedure.
- Re-estimate the model with the B replicative bootstrap data of the selected predictors, and then use the estimated parameter as well as the original data of predictors to forecast the realized volatility;
- The final forecast is derived by averaging these bootstrapped forecasts

For the SSVS-HAR, we use the full hierarchical specification of the SSVS prior according to George & McCulloch (1993) and in a slightly different context according to Korobilis (2013) as follows:

$$p(\beta_i | \xi_i) \sim (1 - \xi_i)N(0, \tau_0^2) + \xi_i N(0, \tau_1^2),$$

$$P(\xi_i | \pi) \sim \text{Bernoulli}(\pi_i).$$

These priors are specified with $\pi = 0.5$, $\tau_0 = 0.001$ and $\tau_1 = 4$. Given these priors, the posterior results of the parameters β, ξ , and σ^2 are sampled with the Gibbs sampling method based on their conditional posterior distributions.

Although LASSO and the BAG approach reduce the dimensionality of the predictor set, BMA and SSVS use a weighting of predictors to select and emphasize the most important variables. The above selection methods are applied to Eq. (4) and the information class-model combinations of (5). This includes the baseline HAR model and necessarily the HAR-RV components in each HAR extension to allow for a deselection of endogenous variance measures.

2.3. Forecast Evaluation

We measure the out-of-sample performance for three different forecast horizons, i.e. short-term (1-day ahead, $h = 1$), medium-term (1-week ahead, $h = 5$) and long-term (1-month ahead, $h = 22$). From the full data set, we used 2/3 as the in-sample or initial training set and the remainder as the out-of-sample set.

The 1-day ahead forecasts are obtained by re-estimating the models on an expanding training set. For the 5- and 22-day ahead forecasts, we follow Corsi (2009) and Corsi & Renò (2012) and estimate the aggregated or averaged realized volatility. Alternatively, one could also forecast h -steps ahead by iterating the model (Marcellino, 2006). However, this forecast method is seen to be less favorable given the propagation of forecast errors due to iterating 1-day ahead forecasts (Sizova, 2011, Ederington & Guan, 2010).

In what follows, we evaluate the set of combinations of the HAR model with a variable selection approach and different sets of exogenous variables presented in Section 2, both with respect to forecast precision and economic significance.

2.3.1. Statistical evaluation

We evaluate the forecast accuracy by the mean squared error (MSE) and mean absolute error (MAE) loss functions, that read

$$MSE = T^{-1} \sum_{t=1}^T \left(RV_t - \widehat{RV}_t \right)^2,$$

$$MAE = T^{-1} \sum_{t=1}^T \left| RV_t - \widehat{RV}_t \right|,$$

where RV_t is the actual realized variance and $\widehat{RV}_t$ is the predicted realized variance. Here, T refers to the number of observations out of sample. We then statistically test whether a model outperforms its peers, by employing the Model Confidence Set (MCS) by Hansen et al. (2011).³

In addition, we use the out-of-sample R_{OS}^2 to measure the proportion of variance explained by the forecasts and to overcome the scale-dependent drawbacks for the loss functions above (Blair et al., 2001, Campbell & Slack, 2008). The R_{OS}^2 is computed as:

$$R_{OS}^2 = 1 - \frac{\sum_{t=1}^T \left(\widehat{RV}_t - RV_t \right)^2}{\sum_{t=1}^T \left(RV_t - \overline{RV} \right)^2},$$

and can be compared to the well-known in-sample R^2 .

2.3.2. Economic evaluation

In order to assess the economic relevance of the forecasts, we calculate the economic gain of the variable selection approaches over the standard AR(1) models. Particularly, we consider an investor who has the mean-variance utility:

$$U(R_p) = E(R_p) - \frac{1}{2} \gamma \text{Var}(R_p)$$

³From our full set of models $M_0 = \{M^i, i = 1, \dots, k\}$, the procedure determines a set of models with superior forecast performance $\widehat{M}_{1-\alpha}^*$. Given a confidence level α , the MCS $\widehat{M}_{1-\alpha}^*$ includes all models from M_0 that are statistically indistinguishable from the best model in the set, i.e. the model with the lowest MSE or MAE, respectively. We implement the procedure using the T_R statistic, the stationary bootstrap with 10 000 draws, and α levels of 0.1 and 0.25.

where γ is the risk-aversion rate. The return of the investor's portfolio is denoted R_p , $E(R_p)$ is the expected portfolio return, and $Var(R_p)$ is the portfolio variance.

The risk-averse investor allocates a budget to a portfolio comprising a risky asset and a risk-free asset. Following Campbell & Thompson (2008), Ferreira & Santa-Clara (2011), and Neely et al. (2014), the optimal portfolio weights, based on information at time t for allocations at time $t + 1$, should be:

$$\hat{w}_t = \frac{1}{\gamma} \left(\frac{\hat{r}_{t+1}}{\hat{\sigma}_{t+1}^2} \right),$$

where $\hat{r}_{t+1}$ is the forecasted excess return of the oil futures based on a moving average with a 1-year (256 trading days) rolling window. The predicted variance $\hat{\sigma}_{t+1}^2$ is derived from the various combinations of the AR and HAR models and the variable selection methods presented in Section 2. In particular, we calculate the portfolio weights at the end of each month, using the predicted next-month variance from the various 22-day ahead models. We repeat the process until the end of our out-of-sample period. Then, the portfolio return at time $t + 1$ is given by:

$$R_{p,t+1} = \hat{w}_t r_{t+1} + R_{t+1}^f,$$

where r_{t+1} and R_{t+1}^f are the excess return of the WTI futures and the risk-free return of a 5-year U.S. treasury note, respectively.

To assess the economic value, we follow Fleming et al. (2001) and Fleming et al. (2003). Based on the the quadratic utility

$$U(r_{p,t}, \gamma) = (1 + r_{p,t}) - \frac{\gamma}{2(1 + \gamma)} (1 + r_{p,t})^2$$

with risk aversion γ and portfolio excess return $r_{p,t}$, we determine the economic value with the constant Δ between two portfolios such that

$$\sum_{t=T_1+1}^T U(r_{p_1,t}) = \sum_{t=T_1+1}^T U(r_{p_2,t} - \Delta).$$

The greater Δ is, the more return a risk-averse investor is willing to sacrifice to switch from portfolio p_1 , representing a benchmark AR(1) model, to p_2 being a HAR/variable selection combination. We follow the literature and use two levels of risk aversion rate for the investor, the mild risk aversion rate $\gamma = 1$ and the strong risk aversion rate $\gamma = 10$.

3. Sample and Data

In this study, we use high-frequency intra-day data of WTI futures in 5-minute resolution. We obtain the data from the Tick Data Database. We choose the most liquid crude oil futures contract instead of a fixed maturity to avoid any price and volatility distortion due to rollover processes.

Following Degiannakis & Filis (2017), we include five different classes of exogenous factors. The first group referring to equity market volatility comprises indices for the U.S. (S&P 500 and DJI), Germany (DAX30), France (CAC40), UK (FTSE), Hong Kong (HSI), Japan (Nikkei225), China (SSEC), as well as the European stock index (STOXX50). The second group includes sentiment and uncertainty measures of financial markets, such as the Global Economic Policy Uncertainty index (GEPU, Baker et al., 2016), S&P500 implied volatility (VIX), oil price implied volatility (OVX), and the BofA Merrill Lynch Global Financial Stress Index (GFSI). The third and fourth groups contain the volatility of commodity futures traded at the CME such as Natural Gas, Gold, Corn, and Soybeans as well as foreign exchange rates with reference U.S. Dollar against the Euro, Japanese Yen, Chinese Yuan, Russian Ruble, and the UK Pound. The volatility of commodities and FX is measured with daily squared returns.

The realized volatilities of the equity indices of nine major stock markets are obtained from the Oxford-Man *Realized Library* (Heber et al., 2009). The remaining lower-frequency data are acquired from Datastream. The sample spans the time period from January 5, 2010 to May 11, 2018. By data pre-processing and synchronization of the data sets, we obtain $n = 2596$ daily observations. To capture market attention with regard to oil, we follow Afkhami et al. (2017) and include Google search volumes for relevant search terms as exogenous variables. Table A.1 in the Appendix provides an overview and

describes the construction.

The summary statistics and the preliminary analysis are given in Table 1. For the realized volatilities of the WTI futures prices, we find the mean and standard deviation to be higher than those of the realized volatilities of the financial market. This underlines the observation that prices in the crude oil futures markets are much more volatile than equity indices. Most importantly, we observe highly significant auto-correlations of realized volatilities for the WTI and for all stock market indices. This reinforces our decision to apply heterogeneous autoregressive frameworks as in the HAR model in both realized volatility and exogenous factors. The only exception to this present auto-correlation in realized volatility are Corn futures, where neither 5, 10, nor 22 lags yield sufficient evidence regarding dependencies between daily realized volatilities. Corn and Natural Gas have the highest mean and standard deviation across all volatility measures. This translates to very high intra-day returns, possibly triggered by intra-day jumps, which is also visible in Figure B.2, plot (b). All volatility measures are found to be stationary, as expected from the construction in Eq. (1). The resolution of Google search volume is expressed in index points. Hence, the summary statistics cannot be directly compared. More importantly for our methodological framework, we find highly significant auto-correlations, which we utilize in the employed autoregressive structure of the HAR extensions.

The visualizations of the remainder of the variables are found in the Appendix. Figure 1 visualizes the daily realized volatility of WTI futures over our data range. Figure B.1 analogously visualizes the volatilities of the main stock market indices. We observe a clustering of highly volatile periods in oil markets for 2011 and 2015-2017. These volatility clusters are also present for equity indices, indicating some degree of connectedness which is explicitly modeled in our model framework. Periods of high and low volatility are also depicted by specific sentiment indices such as the VIX for equity markets (S&P500) and the OVX (WTI) visualized in Figure B.2 (a). Highly volatile in general, Natural Gas, Gold, Corn, and Soybean futures (see Figure B.2 (b)) also show clustering of high- and low-volatility regimes albeit of much higher magnitude and frequency than WTI and equity futures in our sample. The volatility of the FX markets is depicted in Figure B.3 and

Table 1: Summary statistics and preliminary analyses of all variables with $n = 2\,596$ observations spanning an observation period from January 5, 2010 to May 11, 2018.

	Mean	StD	Skew	Kurt	LB $Q(5)$	LB $Q(10)$	LB $Q(22)$	ADF
Panel A: Oil Price Volatility								
WTI	1.6437	2.1386	5.9846	72.07	1328.90***	2154.52***	3902.56***	-22.79***
Panel B: Stock Market Volatility								
DJI	0.6906	1.7313	20.2846	638.60	677.82***	915.44***	1172.76***	-28.99***
CAC40	1.0420	1.4011	6.4217	68.54	3007.06***	4644.45***	6735.51***	-16.00***
FTSE	0.8330	1.7036	19.2349	584.26	629.27***	937.99***	1313.33***	-29.36***
DAX30	1.0383	1.4757	6.4259	68.48	3390.22***	5270.25***	8212.72***	-15.70***
HSI	0.5594	0.7255	8.4951	109.85	968.97***	1338.56***	1836.23***	-21.77***
NIK225	0.7241	1.4659	9.6691	136.76	830.84***	1125.37***	1358.30***	-25.09***
S&P500	0.6808	1.4255	11.5266	232.55	1476.86***	2132.20***	3010.69***	-23.51***
SSEC	1.3308	2.8446	7.5797	79.72	3647.71***	5241.38***	8480.71***	-18.81***
ESTOXX	1.2290	1.9335	12.4211	283.72	1456.65***	2210.27***	3167.91***	-22.29***
Panel C: Commodity Market Volatility								
Natural Gas	7.3079	15.1738	6.7816	78.48	101.62***	150.39***	245.00***	-35.78***
Gold	1.0719	3.2044	15.7551	399.87	43.02***	66.20***	124.05***	-37.66***
Corn	3.0175	14.3666	34.4336	1416.36	1.42	3.87	5.37	-43.39***
Soybeans	1.9011	5.9385	16.4822	402.11	12.74**	25.52***	70.26***	-41.38***
Panel D: Currency Market Volatility								
USD Ind	0.2154	0.4035	5.0454	43.70	39.98***	75.60***	142.44***	-34.35***
EUR/USD	0.3441	0.6227	3.8660	23.73	53.34***	110.52***	221.31***	-34.23***
YEN/USD	0.3815	0.9172	6.8758	67.78	29.96***	52.48***	101.66***	-36.96***
YUAN/USD	0.0218	0.0992	20.0454	577.77	152.56***	181.70***	209.26***	-34.38***
RUB/USD	1.1339	7.4884	25.6058	751.43	872.65***	1046.20***	1174.03***	-23.73***
GBP/USD	0.3301	1.6039	37.3447	1585.73	101.01***	111.08***	115.10***	-37.47***
Panel E: Sentiment Variables								
GFSI	86.6074	8.2043	0.3545	1.65	10590.41***	21029.28***	45525.06***	0.67
VIX	17.1377	5.9161	1.6643	6.36	8783.62***	15907.39***	29015.25***	-2.18**
OVX	33.3028	10.6372	0.8036	3.78	9888.71***	18927.99***	37680.58***	-1.29
EPU	105.9849	62.8497	1.5436	6.41	3500.21***	6109.56***	11052.43***	-10.19***
Panel F: Google Search Volume								
G1: 'Oil Production'	16.8724	11.5640	1.9246	10.54	2415.77***	4227.39***	7041.84***	-13.36***
G2: 'Financial Crisis'	28.0232	12.8161	1.1214	4.89	7446.00***	13463.94***	22717.45***	-4.96***
G3: 'Oil Demand'	12.2505	13.1789	1.3274	5.76	770.13***	1373.19***	2242.12***	-22.37***
G4: 'Oil Price'	14.8950	13.7390	2.0530	8.77	9068.64***	16746.01***	31571.58***	-4.26***
G5: 'OPEC Conference'	2.8353	7.9100	4.6590	36.03	171.30***	287.13***	434.10***	-35.15***

again we find highly volatile markets in 2015-2017. Some important events, such as the sanctions of the U.S. against Russia and China as well as the oil price collapse 2014-2015 are visible as increasingly volatile rate movements in FX markets. Interestingly, this also aligns with Google search queries, featured as indices in Figure B.4, for “oil production” (G1), “oil demand” (G3), and “oil price” (G4) showing that there is an increase in general interest in those topics that we utilize as an additional measure of attention.

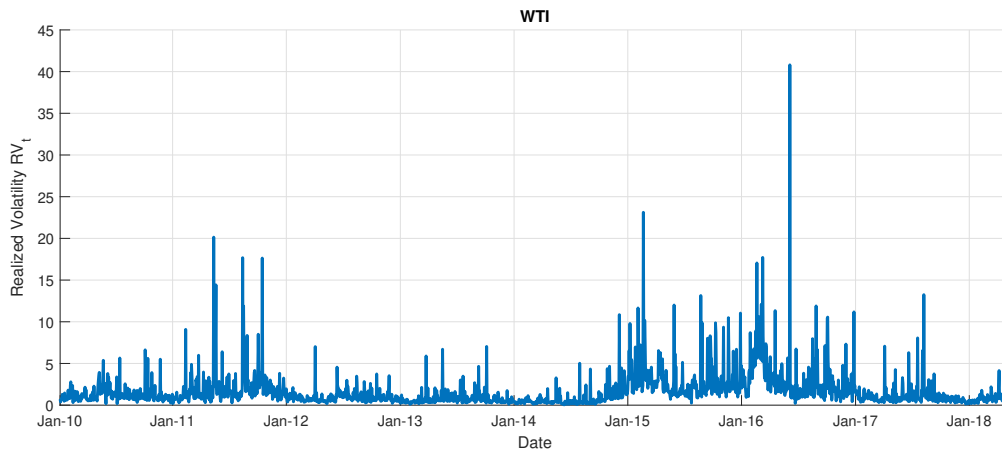


Figure 1: Realized volatility of WTI futures prices on daily resolution from January 5, 2010 to May 11, 2018, calculated based on Eq. (1).

4. Out-of-sample Results

4.1. Forecasting Accuracy

This section presents the results of the out-of-sample analysis. In particular, we forecast the WTI volatility 1-, 5-, and 22-days ahead using the above-mentioned models, methods, and predictors. We stress the fact that all information is known at the time to forecast the next periods volatility (real time).

We start our analysis by assessing the statistical precision of the different models. The results for 1-, 5-, and 22-days ahead forecasts are presented in Table 2 to 4.

For the 1-day ahead forecasts, the best models are BAG-HAR-Google and LASSO-HAR-All in terms of MSE and MAE, respectively. The 25% MCS for the MSE includes all models, but most of the AR models. Thus, the performance of any other model cannot be differentiated from the BAG-HAR-Google. For the MAE, the MCS' are more exclusive. Only four models belong to the 25% MCS (BMA- and BAG-HAR-Google, LASSO- and BAG-HAR-All) and an additional two models (BMA-HAR and BMA-HAR-PCA) are elements of 10% MCS.

Turning to the 5-days ahead forecast, we find a similar pattern in variable selection. The 10% MCS for the MSE loss function is quite large. However, only four models are included in the 25% MCS: LASSO-HAR, LASSO-HAR-Sentiment, LASSO-HAR-Commodity, and LASSO-HAR-FX. The latter is also the best model in terms of the

lowest MSE and MAE. For the MAE, only five models are part of the 10% and 25% MCS. These models include the aforementioned LASSO models as well as AR(1)-All.

The statistical accuracy for 22-days ahead forecasts is almost equal over the entire set of models. At least the MCS includes all of them for both loss functions. The MSE set even includes all variants of the HAR model in the more restrictive 25% set. For MAE, with some other models, all variants of LASSO belong to the 25% MCS. We also find that the best models for both loss functions are the Google variants.

In contrast to Degiannakis & Filis (2017), we cannot see a clear outperformance of HAR models incorporating other asset classes channels over standard HAR models. For all three forecasting horizons, the HAR and the various HAR-X models are either together in the MCS or not. Thus, we conclude that the machine learning approaches, on top of the combination of HAR model and information class, is what makes the difference.

Summarizing, we find an outperformance of the LASSO-HAR variants which are most of the time included in the even more restrictive 25% confidence sets. The very simple AR models are only included on two occasions in the most restrictive sets, indicating the value of the long-term components of the HAR models. The LASSO-HAR-All model, which selects out of a set of all information, is included in the 25% MCS for 1- and 22-days ahead forecast and thus cannot be distinguished from the best performing models in terms of forecast accuracy. We will use this model in the following section to investigate the time-varying selection of variables.

4.2. Time-Varying Drivers

In order to understand the time-variation of different information classes, we investigate the selection of variables over our entire out-of-sample period. In particular, we want to understand 1) which are the most important variables (and at which frequency average) when making no prior choice, 2) how sticky are those variables, and 3) can other information classes replace the own realized volatility (i.e. the WTI RV at 1, 5, and 22-days average).

The top plot in Figure 2 displays the 1-day ahead forecasts variable selection with the LASSO-HAR-All over the whole forecast period. All three RV horizons (red) are selected

Table 2: Statistical evaluations of various volatility models for 1-step forecasts

Model	MSE		MAE		R_{OS}^2
AR(1)	3.3568	(0.1558)	1.1461	(0.0125)	0.5492
HAR	2.7912	(0.5952)	1.0142	(0.0938)	0.6251
LASSO-HAR	3.0235	(0.3811)	1.0271	(0.0938)	0.5939
BMA-HAR	2.9122	(0.3811)	1.0088	(0.2217)	0.6089
BAG-HAR	2.7914	(0.5952)	1.0142	(0.0938)	0.6251
SSVS-HAR	2.8062	(0.3811)	1.0075	(0.0938)	0.6231
AR(1)-Equity	3.8988	(0.1558)	1.1875	(0.0938)	0.4764
HAR-Equity	2.9388	(0.3811)	1.0413	(0.0938)	0.6053
LASSO-HAR-Equity	2.9864	(0.3811)	1.0215	(0.0938)	0.5989
BMA-HAR-Equity	5.5407	(0.3811)	1.0915	(0.0938)	0.2559
BAG-HAR-Equity	2.7979	(0.3811)	1.0019	(0.0988)	0.6242
SSVS-HAR-Equity	2.8050	(0.3811)	1.0078	(0.0938)	0.6233
AR(1)-Sentiment	3.3205	(0.1558)	1.1425	(0.0125)	0.5540
HAR-Sentiment	2.7944	(0.4838)	1.0040	(0.0938)	0.6247
LASSO-HAR-Sentiment	2.9876	(0.3811)	1.0217	(0.0938)	0.5987
BMA-HAR-Sentiment	2.7981	(0.5952)	1.0024	(0.0938)	0.6242
BAG-HAR-Sentiment	2.7868	(0.5952)	1.0080	(0.0938)	0.6257
SSVS-HAR-Sentiment	2.8070	(0.3811)	1.0073	(0.0938)	0.6230
AR(1)-Commodity	3.3389	(0.1558)	1.1389	(0.0091)	0.5516
HAR-Commodity	2.8099	(0.3811)	1.0211	(0.0938)	0.6226
LASSO-HAR-Commodity	3.0269	(0.3811)	1.0253	(0.0938)	0.5935
BMA-HAR-Commodity	2.7984	(0.4838)	1.0139	(0.0938)	0.6242
BAG-HAR-Commodity	2.7975	(0.3811)	1.0150	(0.0938)	0.6243
SSVS-HAR-Commodity	2.8054	(0.3811)	1.0078	(0.0938)	0.6232
AR(1)-FX	3.3670	(0.1558)	1.1546	(0.0325)	0.5478
HAR-FX	3.2294	(0.3811)	1.0521	(0.0938)	0.5663
LASSO-HAR-FX	2.9660	(0.3811)	1.0187	(0.0938)	0.6017
BMA-HAR-FX	2.7426	(0.5952)	1.0131	(0.0938)	0.6317
BAG-HAR-FX	2.7843	(0.5952)	1.0028	(0.0953)	0.6261
SSVS-HAR-FX	2.8035	(0.3811)	1.0075	(0.0938)	0.6235
AR(1)-Google	3.3051	(0.1558)	1.1427	(0.0325)	0.5561
HAR-Google	2.7542	(0.5952)	1.0023	(0.0953)	0.6301
LASSO-HAR-Google	2.9671	(0.3811)	1.0161	(0.0938)	0.6015
BMA-HAR-Google	2.7162	(0.5952)	0.9897	(0.5070)	0.6352
BAG-HAR-Google	2.7075	(1.0000)	0.9897	(0.5070)	0.6364
SSVS-HAR-Google	2.8041	(0.3811)	1.0064	(0.0938)	0.6234
AR(1)-All	3.0871	(0.3811)	1.0954	(0.0938)	0.5854
HAR-All	3.1935	(0.3811)	1.1077	(0.0938)	0.5711
LASSO-HAR-All	2.7787	(0.5952)	0.9734	(1.0000)	0.6268
BMA-HAR-All	2.7324	(0.5952)	1.0898	(0.0325)	0.6330
BAG-HAR-All	2.7337	(0.5952)	0.9822	(0.5070)	0.6328
SSVS-HAR-All	2.7986	(0.4838)	1.0066	(0.0938)	0.6241
AR(1)-PCA	3.1650	(0.1558)	1.0782	(0.0344)	0.5749
HAR-PCA	2.8896	(0.3811)	1.0117	(0.0938)	0.6119
LASSO-HAR-PCA	2.9953	(0.3811)	1.0215	(0.0938)	0.5977
BMA-HAR-PCA	2.8386	(0.3811)	1.0033	(0.1035)	0.6188
BAG-HAR-PCA	2.7992	(0.3811)	1.0085	(0.0938)	0.6240
SSVS-HAR-PCA	2.8047	(0.3811)	1.0070	(0.0938)	0.6233

Note: This table presents the mean squared error (MSE), the mean average error (MAE) loss results and the R_{OS}^2 for short-term forecasts of WTI volatility. Lower values of MSE and MAE loss functions imply higher forecast precision. The corresponding MCS p -values are listed in parentheses on the right sides of the MAE and MSE results, with $p^{MCS} \geq 0.1$ implying an inclusion in the MCS at 10% confidence while $p^{MCS} \geq 0.25$ implies inclusion in the MCS at the 25% confidence level. The R_{OS}^2 measures the proportion of variance explained by the forecasts: $R_{OS}^2 = 1 - \frac{\sum_{t=T_1}^{t=T} (\widehat{RV}_t - RV_t)^2}{\sum_{t=T_1}^{t=T} (RV_t - \overline{RV})^2}$. Higher values of R_{OS}^2 suggest higher forecast precision.

Table 3: Statistical evaluations of various volatility models for 5-step forecasts

Model	MSE		MAE		R_{OS}^2
AR(1)	1.3563	(0.0899)	1.0005	(0.0207)	0.6131
HAR	1.2262	(0.1700)	0.9738	(0.0207)	0.6502
LASSO-HAR	1.0817	(0.5343)	0.9077	(0.4274)	0.6914
BMA-HAR	1.2606	(0.1700)	0.9315	(0.0207)	0.6404
BAG-HAR	1.2276	(0.1352)	0.9744	(0.0207)	0.6498
SSVS-HAR	1.2408	(0.1284)	0.9560	(0.0207)	0.6460
AR(1)-Equity	2.6016	(0.0899)	1.0409	(0.0207)	0.2578
HAR-Equity	1.2215	(0.1700)	0.9654	(0.0207)	0.6515
LASSO-HAR-Equity	1.1642	(0.2327)	0.9438	(0.0207)	0.6679
BMA-HAR-Equity	1.3090	(0.1352)	0.9487	(0.0207)	0.6266
BAG-HAR-Equity	1.2593	(0.1284)	0.9918	(0.0182)	0.6408
SSVS-HAR-Equity	1.2350	(0.0899)	0.9553	(0.0207)	0.6477
AR(1)-Sentiment	1.3264	(0.0899)	0.9941	(0.0207)	0.6216
HAR- Sentiment	1.2425	(0.1352)	0.9835	(0.0207)	0.6455
LASSO-HAR-Sentiment	1.0913	(0.5343)	0.9107	(0.4274)	0.6887
BMA-HAR-Sentiment	1.2521	(0.0899)	0.9577	(0.0207)	0.6428
BAG-HAR-Sentiment	1.2337	(0.1352)	0.9781	(0.0207)	0.6480
SSVS-HAR-Sentiment	1.2411	(0.1284)	0.9561	(0.0207)	0.6459
AR(1)-Commodity	1.3505	(0.0899)	0.9982	(0.0207)	0.6147
HAR-Commodity	1.2310	(0.1352)	0.9788	(0.0207)	0.6488
LASSO-HAR-Commodity	1.0835	(0.5343)	0.9087	(0.4274)	0.6909
BMA-HAR-Commodity	1.2441	(0.1284)	0.9525	(0.0207)	0.6451
BAG-HAR-Commodity	1.2291	(0.1352)	0.9751	(0.0207)	0.6494
SSVS-HAR-Commodity	1.2355	(0.1284)	0.9557	(0.0207)	0.6475
AR(1)- FX	1.3719	(0.0899)	1.0043	(0.0207)	0.6086
HAR- FX	1.1413	(0.1700)	0.9484	(0.0207)	0.6744
LASSO-HAR-FX	1.0624	(1.0000)	0.8946	(1.0000)	0.6969
BMA-HAR-FX	1.1952	(0.1700)	0.9312	(0.0207)	0.6590
BAG-HAR-FX	1.2697	(0.0899)	1.0078	(0.0068)	0.6378
SSVS-HAR-FX	1.2347	(0.0899)	0.9541	(0.0207)	0.6478
AR(1)-Google	1.2448	(0.1700)	0.9864	(0.0207)	0.6449
HAR-Google	1.3549	(0.0899)	0.9978	(0.0207)	0.6135
LASSO-HAR-Google	1.1934	(0.1700)	0.9356	(0.0207)	0.6595
BMA-HAR-Google	1.3016	(0.0899)	0.9647	(0.0207)	0.6287
BAG-HAR-Google	1.3103	(0.0899)	0.9900	(0.0207)	0.6262
SSVS-HAR-Google	1.2424	(0.0899)	0.9563	(0.0207)	0.6456
AR(1)-All	1.4600	(0.0899)	0.9527	(0.0207)	0.5835
HAR-All	1.2649	(0.1284)	0.9780	(0.0207)	0.6392
LASSO-HAR-All	1.1733	(0.1700)	0.9527	(0.0207)	0.6653
BMA-HAR-All	1.2582	(0.1352)	0.9513	(0.0207)	0.6411
BAG-HAR-All	1.2898	(0.0899)	0.9944	(0.0182)	0.6320
SSVS-HAR-All	1.2140	(0.1284)	0.9559	(0.0207)	0.6537
AR(1)-PCA	1.1660	(0.2327)	0.9162	(0.4274)	0.6674
HAR-PCA	1.2578	(0.1284)	0.9969	(0.0120)	0.6412
LASSO-HAR-PCA	1.0958	(0.2327)	0.9202	(0.0207)	0.6874
BMA-HAR-PCA	1.2570	(0.0899)	0.9642	(0.0207)	0.6414
BAG-HAR-PCA	1.2486	(0.1284)	0.9921	(0.0182)	0.6438
SSVS-HAR-PCA	1.2409	(0.1284)	0.9562	(0.0207)	0.6460

Note: This table presents the mean squared error (MSE), the mean average error (MAE) loss results, and the R_{OS}^2 for medium-term forecasts of WTI volatility. Lower values of MSE and MAE loss functions imply higher forecast precision. The corresponding MCS p -values are listed in parentheses on the right side of the MAE and MSE results, with $p^{MCS} \geq 0.1$ implying an inclusion in the MCS at 10% confidence while $p^{MCS} \geq 0.25$ implies inclusion in the MCS at the 25% confidence level. The R_{OS}^2 measures the proportion of variance explained by the forecasts: $R_{OS}^2 = 1 - \frac{\sum_{t=T_1}^{t=T} (\widehat{RV}_t - RV_t)^2}{\sum_{t=T_1}^{t=T} (RV_t - \overline{RV})^2}$. Higher values of R_{OS}^2 suggest higher forecast precision.

Table 4: Statistical evaluations of various volatility models for 22-step forecasts

Model	MSE		MAE		R^2_{OS}
AR(1)	0.9316	(0.2538)	0.9087	(0.0743)	0.6040
HAR	0.7596	(0.3160)	0.7691	(0.6133)	0.6771
LASSO-HAR	0.7127	(0.5850)	0.7536	(0.8506)	0.6970
BMA-HAR	0.7494	(0.5850)	0.7831	(0.8131)	0.6814
BAG-HAR	0.7612	(0.3160)	0.7701	(0.1741)	0.6764
SSVS-HAR	0.7413	(0.3160)	0.7851	(0.1741)	0.6849
AR(1)-Equity	3.3078	(0.0586)	1.0013	(0.1741)	-0.4061
HAR-Equity	0.7240	(0.5850)	0.7686	(0.1741)	0.6922
LASSO-HAR-Equity	0.7356	(0.3779)	0.7607	(0.8506)	0.6873
BMA-HAR-Equity	0.7413	(0.3779)	0.7835	(0.1741)	0.6849
BAG-HAR-Equity	0.7353	(0.3779)	0.7665	(0.6133)	0.6875
SSVS-HAR-Equity	0.7405	(0.3160)	0.7729	(0.1741)	0.6852
AR(1)-Sentiment	0.8986	(0.3160)	0.9041	(0.1575)	0.6180
HAR- Sentiment	0.7556	(0.3160)	0.7697	(0.6133)	0.6788
LASSO-HAR-Sentiment	0.7245	(0.5850)	0.7575	(0.8506)	0.6920
BMA-HAR-Sentiment	0.7318	(0.3779)	0.7813	(0.1741)	0.6889
BAG-HAR-Sentiment	0.7594	(0.3160)	0.7714	(0.1741)	0.6772
SSVS-HAR-Sentiment	0.7411	(0.3160)	0.7851	(0.1741)	0.6850
AR(1)-Commodity	0.9407	(0.2520)	0.9121	(0.0655)	0.6001
HAR-Commodity	0.7733	(0.3160)	0.7751	(0.1741)	0.6713
LASSO-HAR-Commodity	0.7406	(0.3160)	0.7626	(0.8131)	0.6852
BMA-HAR-Commodity	0.7503	(0.3160)	0.7824	(0.1741)	0.6811
BAG-HAR-Commodity	0.7743	(0.3160)	0.7750	(0.1741)	0.6709
SSVS-HAR-Commodity	0.7268	(0.3779)	0.7724	(0.6133)	0.6911
AR(1)- FX	0.9652	(0.0586)	0.8985	(0.1741)	0.5897
HAR- FX	0.7704	(0.3160)	0.7773	(0.1741)	0.6725
LASSO-HAR-FX	0.7235	(0.5850)	0.7575	(0.8506)	0.6925
BMA-HAR-FX	0.7451	(0.3160)	0.7900	(0.1741)	0.6833
BAG-HAR-FX	0.7722	(0.3160)	0.7754	(0.1741)	0.6718
SSVS-HAR-FX	0.7457	(0.3160)	0.7752	(0.1741)	0.6830
AR(1)-Google	0.8153	(0.3160)	0.8919	(0.1741)	0.6535
HAR-Google	0.6851	(0.9752)	0.7476	(1.0000)	0.7088
LASSO-HAR-Google	0.7228	(0.5850)	0.7570	(0.8506)	0.6928
BMA-HAR-Google	0.6811	(1.0000)	0.7595	(0.8506)	0.7105
BAG-HAR-Google	0.6885	(0.5850)	0.7492	(0.8506)	0.7074
SSVS-HAR-Google	0.7369	(0.3779)	0.7836	(0.1741)	0.6867
AR(1)-All	1.7713	(0.0586)	0.9155	(0.1741)	0.2471
HAR-All	0.7471	(0.3160)	0.7786	(0.1741)	0.6824
LASSO-HAR-All	0.7451	(0.3160)	0.7691	(0.6133)	0.6833
BMA-HAR-All	0.7245	(0.5850)	0.7766	(0.6133)	0.6920
BAG-HAR-All	0.7469	(0.3160)	0.7687	(0.6133)	0.6825
SSVS-HAR-All	0.7426	(0.3779)	0.7617	(0.8506)	0.6843
AR(1)-PCA	0.8703	(0.3160)	0.8518	(0.1741)	0.6300
HAR-PCA	0.7714	(0.3160)	0.7781	(0.1741)	0.6721
LASSO-HAR-PCA	0.7410	(0.3160)	0.7640	(0.6133)	0.6850
BMA-HAR-PCA	0.7539	(0.3160)	0.7862	(0.1741)	0.6795
BAG-HAR-PCA	0.7842	(0.3160)	0.7827	(0.1741)	0.6667
SSVS-HAR-PCA	0.7418	(0.3160)	0.7855	(0.1741)	0.6847

Note: This table presents the mean squared error (MSE), the mean average error (MAE) loss results, and the R^2_{OS} for long-term forecasts of the volatility of WTI. Lower values of MSE and MAE loss functions imply higher forecast precision. The corresponding MCS p -values are listed in parentheses on the right side of the MAE and MSE results, with $p^{MCS} \geq 0.1$ implying an inclusion in the MCS at the 10% confidence level while $p^{MCS} \geq 0.25$ implies an inclusion in the MCS at the 25% confidence level. The R^2_{OS} measures the proportion of variance explained by the forecasts: $R^2_{OS} = 1 - \frac{\sum_{t=T_1}^{t=T} (\bar{RV}_t - RV_t)^2}{\sum_{t=T_1}^{t=T} (RV_t - \bar{RV})^2}$. Higher values of R^2_{OS} suggest higher forecast precision. The best model per measure is highlighted in bold.

throughout the out-of-sample period. In addition, we find that the most consistent variables are the OVX (oil price uncertainty) at 5 and 22 days horizon and the Google search volume for “Oil Price” the day before.

The middle plot in Figure 2 shows the 5-days ahead forecast variable selection. We find that the RV of the WTI the day before is never selected. Only the weekly and monthly RV appear to be relevant for the forecast. Other variables are selected at the daily horizon, probably replacing RVs as a more suitable variable for prediction. Although we do not find a persistent variable in the short-term, weekly and monthly averages of the Chinese stock market (SSEC), sentiment (OVX, VIX, EPU), commodities (Corn, Soybean, Natural Gas) and FX (USD, Yen/USD, GBP/USD) are selected almost throughout the entire sample.

A look at the bottom plot of Figure 2 for 22 days ahead reveals that LASSO-HAR-All uses many different variables throughout the sample at different time horizons, e.g. NIKKEI and SSEC, VIX, OVX, Natural Gas, Soybeans, and Google search volumes for “Financial Crisis” and “OPEC Conference”.

With increasing forecast horizon, LASSO identifies an increasing number of exogenous predictors that are useful for volatility forecasting. This ML approach also shows that the selection of variables is varying over time. Apart from the more persistent variables mentioned above, several exogenous variables only play a role for a limited period of time and we observe clustering of variable selections. This culminates in the observation that for the 22-days forecasting horizon, there appears to be a structural break in variable selection at the end of 2016. LASSO is not choosing stock market volatility, which was significant until this date, while some commodity and FX-related variables are integrated in the forecasts. As to why LASSO discards several exogenous predictors at the same time is not within the scope of this work and we leave this question open for future research.

Lastly, we also consider the ranking of exogenous variables with regard to variable importance. We follow Narajewski & Ziel (2020) and calculate the average variable im-

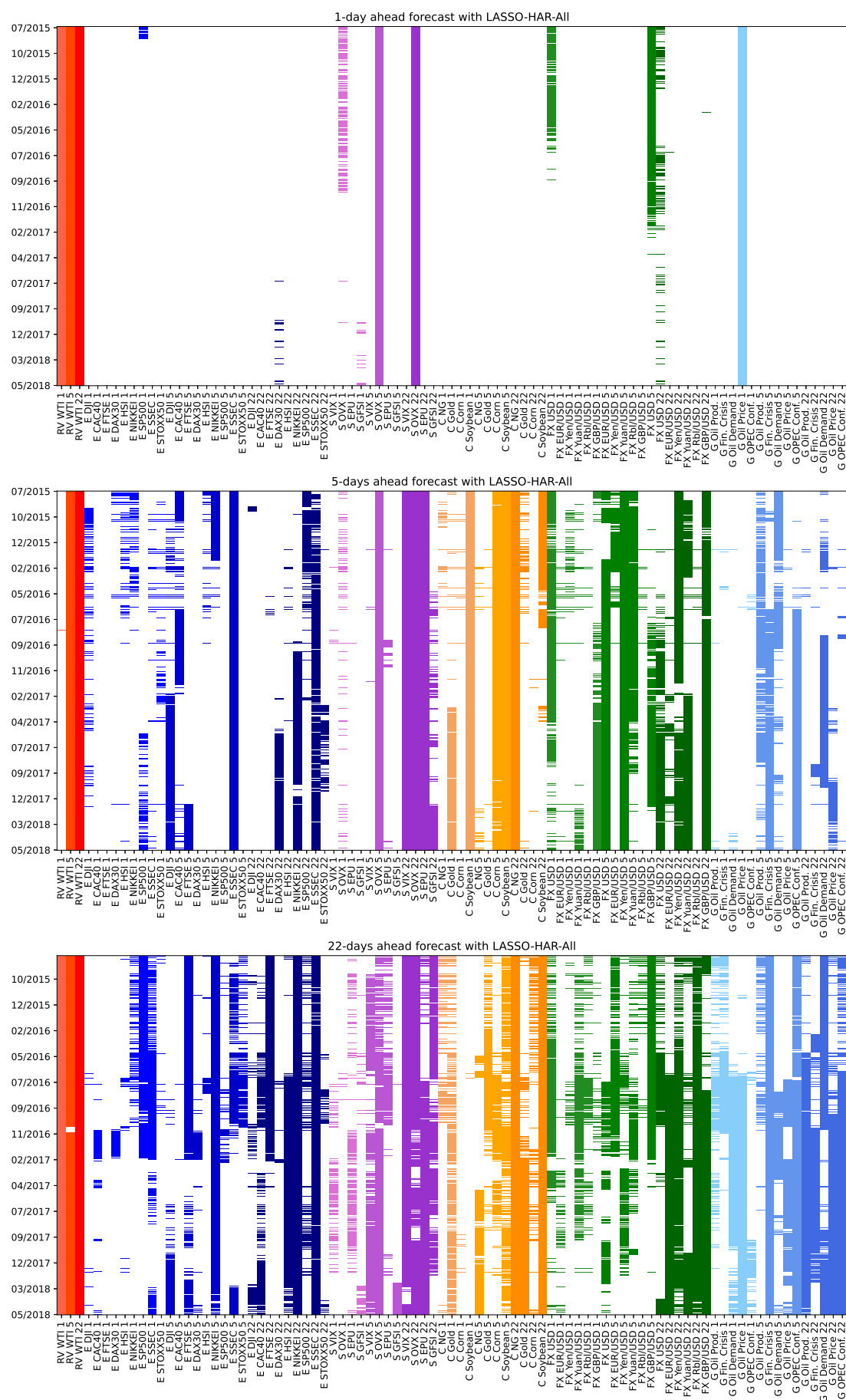


Figure 2: Variable selection over time with the LASSO-HAR-All.

portance over the out-of-sample by

$$\overline{VI}_i = T^{-1} \sum_{t=1}^T VI_{t,i},$$

$$VI_{t,i} = \frac{|\widehat{\beta}_i^{(t)}|}{\sum_{i=1}^N |\widehat{\beta}_i^{(t)}|},$$

where T is the number of out-of-sample observations, N the number of variables, and $\widehat{\beta}_i^{(t)}$ refers to the estimated coefficient of variable i for the out-of-sample at observation t . Note that for the LASSO estimation all variables are standardized. Hence, we are able to compare the individual contribution based on these estimates. Table 5 lists the ten most important variables per forecast horizon.

Rank	1 day	VI	5 days	VI	22 days	VI
1	RV WTI 22	47.8%	RV WTI 22	53.6%	RV WTI 22	66.4%
2	RV WTI 5	30.9%	FX Yuan/USD 22	9.4%	FX EUR/USD 22	5.6%
3	RV WTI 1	8.9%	RV WTI 5	7.4%	FX Yuan/USD 22	4.1%
4	FX USD 5	5.1%	S OVX 22	5.2%	FX Yen/USD 22	2.8%
5	S OVX 22	3.5%	FX Yen/USD 5	4.6%	E NIKKEI 22	2.7%
6	S VIX 5	2.0%	FX Yuan/USD 5	4.3%	E HSI 22	2.6%
7	FX USD 22	0.7%	FX GBP/USD 22	3.0%	S GFSI 22	1.6%
8	FX USD 1	0.6%	FX USD 5	2.7%	RV WTI 5	1.5%
9	G Oil Price 1	0.5%	FX USD 22	2.2%	FX GBP/USD 22	1.3%
10	S OVX 1	<0.1%	FX Yen/USD 22	2.0%	C Soybean 22	1.3%

Table 5: Ranked variable importance of individual variables (top 10) across the applied forecasting horizons.

Although there exists some correlation between the rankings of selections and importance, there are also major differences. We find RVs of the WTI itself always in the ranking of 22-days, 5-days, and 1-day. The most important variable over all forecast horizons is the 22 day RV of the WTI which contributes about 50% to the forecasted RV (or more). We find the FX channel to have a fair share of importance, especially in comparison with the other information channels. Sentiment indices (VIX, OVX, and GFSI) are also among the top 10 for all forecasting horizons. However, in contrast to the frequency of selection, Google search volumes do not carry much variable importance for forecasts beyond 1-day. In fact, the Google attention measures are not even in the top 25.

This observation is also confirmed by looking at the aggregated channel level. In Table 6, we report the sum of the variable importance per variable of a information channel (i.e. per variable and time aggregated). Especially for longer horizons, the contribution of FX is quite substantial. Again, we find little support for Google search volume. We also point out the contrast between the variable importance of the commodity channel and the frequency of selection for commodity variables. The equity channel appears to be become more relevant for longer forecasting horizons, which suggests that equity market volatility does not have an immediate impact but aggregates its influence over time.

Rank	1 day	VI	5 days	VI	22 days	VI
1	RV	87.6%	RV	60.0%	RV	68.2%
2	FX	6.3%	FX	28.8%	FX	17.2%
3	Sentiment	5.6%	Sentiment	6.9%	Equity	8.6%
4	Google	0.5%	Equity	2.2%	Sentiment	3.5%
5	Equity	<0.1%	Commodity	1.5%	Commodity	2.6%
6	Commodity	0.0%	Google	3.0%	Google	1.1%

Table 6: Ranked variable importance of (aggregated) information channels across the applied forecasting horizons.

We summarize that for all forecasting horizons, the realized volatilities of the WTI are the most important variables to predict future realized volatility of the WTI. However, the observed other information channels do contribute to the predictions, and in this way either substitute or add information on top of the RVs. In particular, FX appears to be an important source of information.

4.3. Economic Value

From a statistical point of view, standard HAR models are outperformed by their ML-augmented counterparts. In the next step, we evaluate the economic value of the model augmentations. In particular, we focus on portfolio construction based on volatility forecasts produced by all predictive models. We assess the annualized return which a risk averse investor would sacrifice in order to switch from a benchmark portfolio—AR(1) models without ML-driven variable selection—to any other portfolio, including those that are produced by the augmented models. The results for the different forecast horizons are presented in Table 7.

In contrast to the statistical superiority of the LASSO variants, we observe that this outperformance does not translate into the best economic value. We find that the BMA models are superior and allow a risk averse investor with a quadratic utility function to improve their overall utility. In particular, the BMA-HAR-FX yields the highest annualized return. This model also results in the highest economic value compared to an AR(1) model. An investor with $\gamma = 1$ would be willing to switch from a standard AR(1) model to a BMA-HAR-FX model up to 5.28%. Even to switch from the standard HAR model, the opportunity to incorporate more information than just past realized volatility would be worth more than 1% ($5.28\% - 4.22\%$). However, in risk-adjusted terms, we find that BAG models have the highest Sharpe ratios. The BAG-HAR-All produces the highest Sharpe ratio in our out-of-sample exercise (0.0678). The LASSO variants, which yielded the highest statistical accuracy, perform similarly to the standard HAR models. The variable selection does not appear to be able to translate the higher accuracy to economic performance.

In terms of adding other information channels, we find economic value in incorporating such information in volatility models. Although we did not necessarily find an outperformance on a statistical accuracy, additional information result in higher economic value in the majority of cases. For example, the HAR model has a smaller economic value and Sharpe ratio than its sentiment, commodity, and FX augmented counterparts as well as the HAR-All and HAR-PCA. Similar patterns can be observed between the ML-estimated HAR models and their variants with additional information.

5. Conclusion

This paper demonstrates how extending existing models of realized volatility with additional information from other channels and recent ML techniques benefits the quality of forecasts and subsequent portfolio performance. We focus on modeling realized volatility of the most liquid WTI crude oil futures prices. Existing HAR models (Corsi, 2009) are first enhanced with exogenous factors, which has been shown to improve the forecast performance across different horizons (Ma et al., 2018, Luo et al., 2020). Motivated by

Table 7: Economic evaluations

	$\gamma = 1$				$\gamma = 10$			
	r_p	$\frac{r_p}{\sigma_p}$	Δ_1	Δ_2	r_p	$\frac{r_p}{\sigma_p}$	Δ_1	Δ_2
AR(1)	-1.7566	-0.0338	0.0000	0.0000	1.3663	0.2558	0.0000	0.0000
HAR	2.4220	0.0541	4.2157	4.2157	1.7842	0.3849	0.4402	0.4402
LASSO-HAR	1.9851	0.0443	3.7714	3.7714	1.7405	0.3757	0.3938	0.3938
BMA-HAR	1.3423	0.0243	3.1058	3.1058	1.6762	0.2960	0.3242	0.3242
BAG-HAR	2.4014	0.0536	4.1947	4.1947	1.7821	0.3843	0.4380	0.4380
SSVS-HAR	2.7588	0.0500	4.5450	4.5450	1.8179	0.3203	0.4745	0.4745
AR(1)-Equity	2.3220	0.0434	0.0000	4.1024	1.7742	0.3225	0.0000	0.4283
HAR-Equity	2.1395	0.0467	-0.1733	3.9270	1.7559	0.3706	-0.0180	0.4100
LASSO-HAR-Equity	2.0429	0.0455	-0.2693	3.8299	1.7463	0.3758	-0.0281	0.3999
BMA-HAR-Equity	2.0508	0.0364	-0.2768	3.8225	1.7471	0.3019	-0.0289	0.3990
BAG-HAR-Equity	2.3699	0.0528	0.0596	4.1625	1.7790	0.3827	0.0063	0.4346
SSVS-HAR-Equity	3.1705	0.0605	0.8574	4.9694	1.8590	0.3443	0.0895	0.5188
AR(1)-Sentiment	-1.3490	-0.0253	0.0000	0.4041	1.4071	0.2567	0.0000	0.0422
HAR- Sentiment	2.5591	0.0566	3.9464	4.3549	1.7979	0.3845	0.4120	0.4547
LASSO-HAR-Sentiment	2.1009	0.0469	3.4811	3.8890	1.7521	0.3781	0.3634	0.4060
BMA-HAR-Sentiment	2.7992	0.0506	4.1773	4.5861	1.8219	0.3207	0.4361	0.4788
BAG-HAR-Sentiment	2.5643	0.0567	3.9517	4.3601	1.7984	0.3846	0.4126	0.4552
SSVS-HAR- Sentiment	2.7580	0.0499	4.1355	4.5442	1.8178	0.3202	0.4317	0.4744
AR(1)-Commodity	-2.0925	-0.0410	0.0000	-0.3328	1.3327	0.2539	0.0000	-0.0347
HAR-Commodity	3.0198	0.0653	5.1612	4.8237	1.8440	0.3858	0.5389	0.5037
LASSO-HAR-Commodity	2.4507	0.0539	4.5811	4.2441	1.7871	0.3797	0.4783	0.4431
BMA-HAR-Commodity	3.2071	0.0572	5.3392	5.0016	1.8627	0.3232	0.5574	0.5222
BAG-HAR-Commodity	3.0205	0.0652	5.1617	4.8242	1.8440	0.3848	0.5389	0.5037
SSVS-HAR-Commodity	2.6341	0.0497	4.7581	4.4210	1.8054	0.3307	0.4967	0.4615
AR(1)- FX	-3.7883	-0.0749	0.0000	-2.0091	1.1632	0.2236	0.0000	-0.2096
HAR- FX	2.9859	0.0653	6.8364	4.7896	1.8406	0.3893	0.7138	0.5001
LASSO-HAR-FX	2.2795	0.0506	6.1131	4.0704	1.7699	0.3800	0.6382	0.4250
BMA-HAR-FX	3.4803	0.0596	7.3270	5.2775	1.8900	0.3151	0.7649	0.5510
BAG-HAR-FX	2.8013	0.0615	6.6471	4.6014	1.8221	0.3867	0.6940	0.4804
SSVS-HAR-FX	2.4326	0.0469	6.2610	4.2174	1.7853	0.3344	0.6536	0.4403
AR(1)-Google	-2.5096	-0.0461	0.0000	-0.7515	1.2910	0.2309	0.0000	-0.0784
HAR-Google	2.3959	0.0541	4.9514	4.1897	1.7816	0.3887	0.5170	0.4375
LASSO-HAR-Google	2.0954	0.0468	4.6446	3.8835	1.7515	0.3783	0.4849	0.4055
BMA-HAR-Google	2.7495	0.0507	5.2993	4.5369	1.8169	0.3257	0.5532	0.4736
BAG-HAR-Google	2.5362	0.0571	5.0944	4.3325	1.7956	0.3904	0.5319	0.4524
SSVS-HAR-Google	2.7582	0.0500	5.3068	4.5444	1.8178	0.3205	0.5540	0.4744
AR(1)-All	-0.5690	-0.0110	0.0000	1.1862	1.4851	0.2787	0.0000	0.1237
HAR-All	2.8910	0.0603	3.4924	4.6900	1.8311	0.3697	0.3647	0.4897
LASSO-HAR-All	3.0138	0.0660	3.6203	4.8182	1.8434	0.3899	0.3781	0.5031
BMA-HAR-All	3.2646	0.0596	3.8635	5.0622	1.8685	0.3315	0.4034	0.5285
BAG-HAR-All	3.1723	0.0678	3.7804	4.9789	1.8592	0.3847	0.3948	0.5199
SSVS-HAR-All	2.1335	0.0462	2.7254	3.9204	1.7553	0.3676	0.2846	0.4093
AR(1)-PCA	-1.4749	-0.0280	0.0000	0.2795	1.3945	0.2576	0.0000	0.0292
HAR-PCA	2.7687	0.0608	4.2853	4.5682	1.8189	0.3858	0.4474	0.4770
LASSO-HAR-PCA	2.2566	0.0502	3.7648	4.0472	1.7677	0.3798	0.3931	0.4226
BMA-HAR-PCA	3.0610	0.0550	4.5697	4.8527	1.8481	0.3228	0.4770	0.5066
BAG-HAR-PCA	2.7372	0.0601	4.2532	4.5360	1.8157	0.3851	0.4441	0.4736
SSVS-HAR-PCA	2.7722	0.0502	4.2758	4.5587	1.8192	0.3205	0.4464	0.4759

Note: Results for 22-step ahead forecasts for the economic evaluation. We present the annualized average excess portfolio return r_p , the average portfolio Sharpe Ratio $\frac{r_p}{\sigma_p}$ and the economic value Δ of the portfolio over the standard AR model within the models class. Here, Δ_1 and Δ_2 refer to the economic value compared to the AR(1) model with the information about the asset class and the standard AR(1) model, respectively. The model with the highest economic value is highlighted in bold.

Degiannakis & Filis (2017), we include several different information channels that include major stock markets, relevant FX market pairs, sentiment indices, other linked commodity markets, and Google search volumes for relevant search terms. We then extend these models with four machine learning approaches which pick the most suitable factors to forecast realized volatility over one day, one week, and one month. The set of models is completed by a PCA variant and a model that includes all possible exogenous variables.

We present several novel findings. First and foremost, including ML to choose from a set of exogenous variables improves the quality of realized volatility forecasts. In particular, LASSO variants show significant improvements. Secondly, we find that the variable selection process strongly depends on the forecasting horizon. While for short-term forecasts, endogenous factors dominate the selection of predictors, the number of predictors increases when the forecasting horizon is increased. Sentiment variables, such as the EPU or OVX, realized volatility of other stock markets and commodities as well as FX markets become increasingly important for longer horizons. We also show that a combination of short, medium and long-term averages of text-based Google indicators are relevant exogenous factors that are included in the predictor set generated by ML algorithms. Thirdly, we show that portfolios constructed with ML-HAR variants (particularly BMA) provide higher returns than the baseline AR and HAR models and its extensions with exogenous factors. Surprisingly, forecast accuracy does not necessarily imply the highest portfolio performance. Lastly, we show that the selection process with its dynamic implementation is time-varying with respect to the variable choices. Variable selection mostly is clustered, while these clusters differ across forecasting horizons. In particular, for the stock market channel, the results point towards structural changes, especially for longer time horizons. Although the most important variables remain the past (aggregates) of realized volatility of the WTI, a great deal of information is coming from the FX channel (especially for longer forecast horizons). In contrast, Google search volume and other commodity volatilities do not carry large fractions of variable importance for the forecast of WTI realized volatility.

Future research might focus on as to why these structural changes in the variable

selection process exist and why these changes materialize differently across forecasting horizons. Having shown that superior fit does not necessarily translate to superior portfolio performance in our portfolio selection application, future work could, for example, address the exploitation of ML generated forecasts of realized volatility for other commodity or asset classes or extend our framework to other ML models such as reinforcement learning (see e.g. Lavko et al., 2023).

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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Appendix A. Additional Tables

Table A.1: Constructions of exogenous predictors

Category	Data	Variable construction
Financial market volatility	DJI, CAC40, FTSE, DAX30, HSI, Nikkei225, S&P500, SSE, STOXX50	We obtain the realized volatilities of the 9 stock indices come from Oxford-Man Institute's "realized library"
Sentiment	VIX, OVX, US EPU, GFSI index	We obtain the four indices from the Datastream database. The first difference of the variables is used as the predictors.
Commodity market	Natural gas, Gold, Corn, Soybean	The daily prices of the four commodity futures are obtained from the Wind database. The return is computed by $r_t = 100 \cdot (\log P_t - \log P_{t-1})$. We use the square returns r_t^2 of the four commodity futures as predictors.
Currency market	US dollar index, Euro/US exchange rate, Japanese Yen/US exchange rate, Chinese Yuan/US exchange rate, Russia Rouble/US exchange rate, UK pound/US exchange rate	The daily prices of US dollar index and the five foreign exchange rates are obtained from Wind database. The return is computed by $r_t = 100 \cdot (\log P_t - \log P_{t-1})$. We use the square returns r_t^2 of the US dollar index and the five foreign exchange rates as predictors.
Google search volume	Oil production, Financial crisis, Oil demand, Oil price, OPEC conference	The google search volume indices are downloaded from the Google index database. We employ the first-difference of the google search indices for the key words as the predictors.

Note: VIX and OVX denote the implied volatility index for S&P 500 and WTI, US EPU is the economic policy uncertainty index for the US and GFSI index is the global financial stress index.

Appendix B. Additional Figures

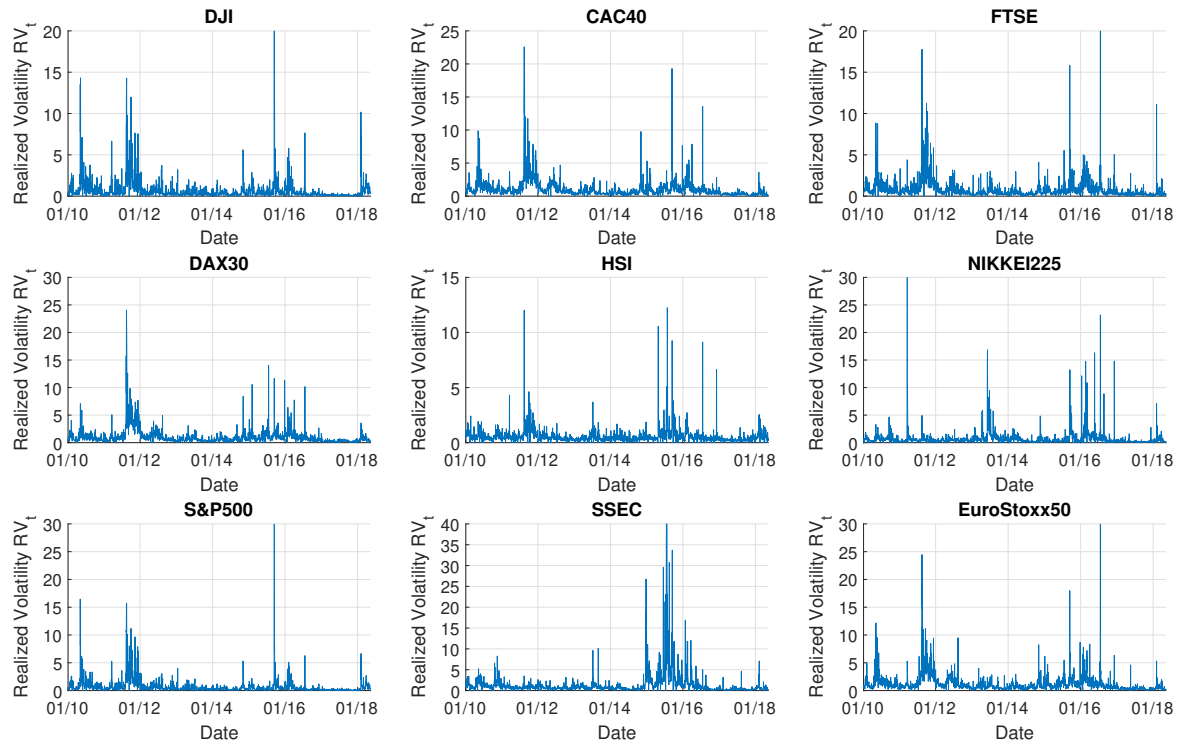


Figure B.1: Realized volatility of stock indices on daily resolution from January 5, 2010 to May 11, 2018, obtained from the Oxford-Man Realized Library.

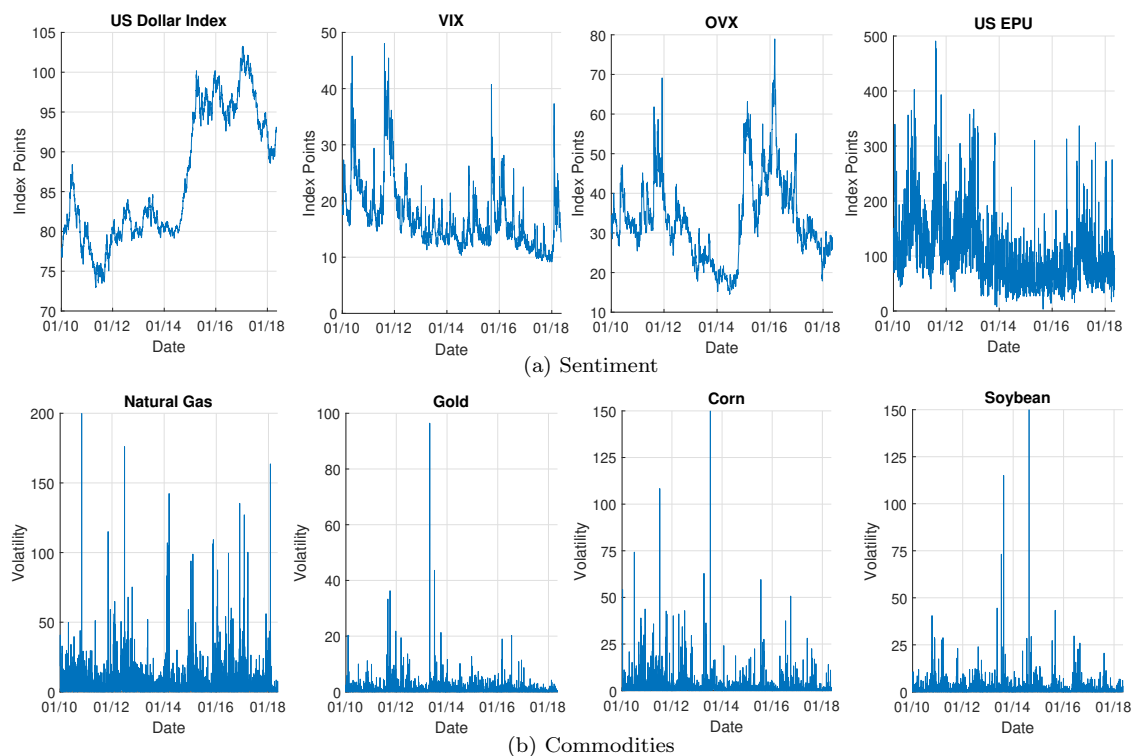


Figure B.2: Sentiment indices as index points and realized volatility of commodity markets, proxied by squared daily returns, on daily resolution from January 5, 2010 to May 11, 2018, obtained from Datastream.

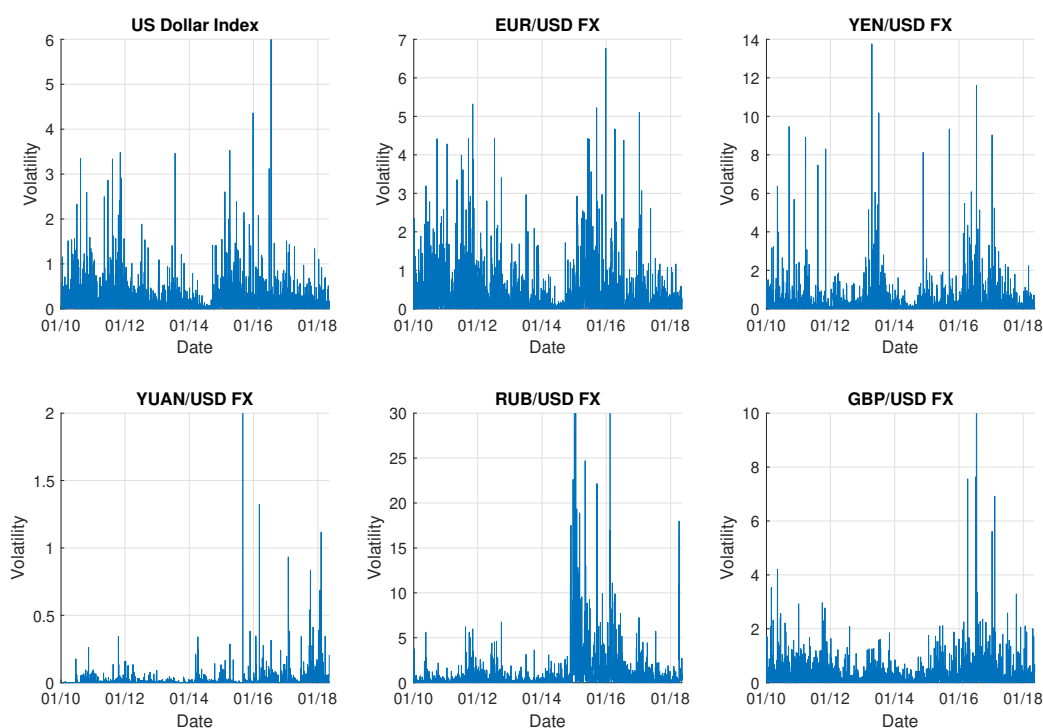


Figure B.3: Realized volatility of FX markets, proxied by squared daily returns, on daily resolution from January 5, 2010 to May 11, 2018, obtained from Datastream.

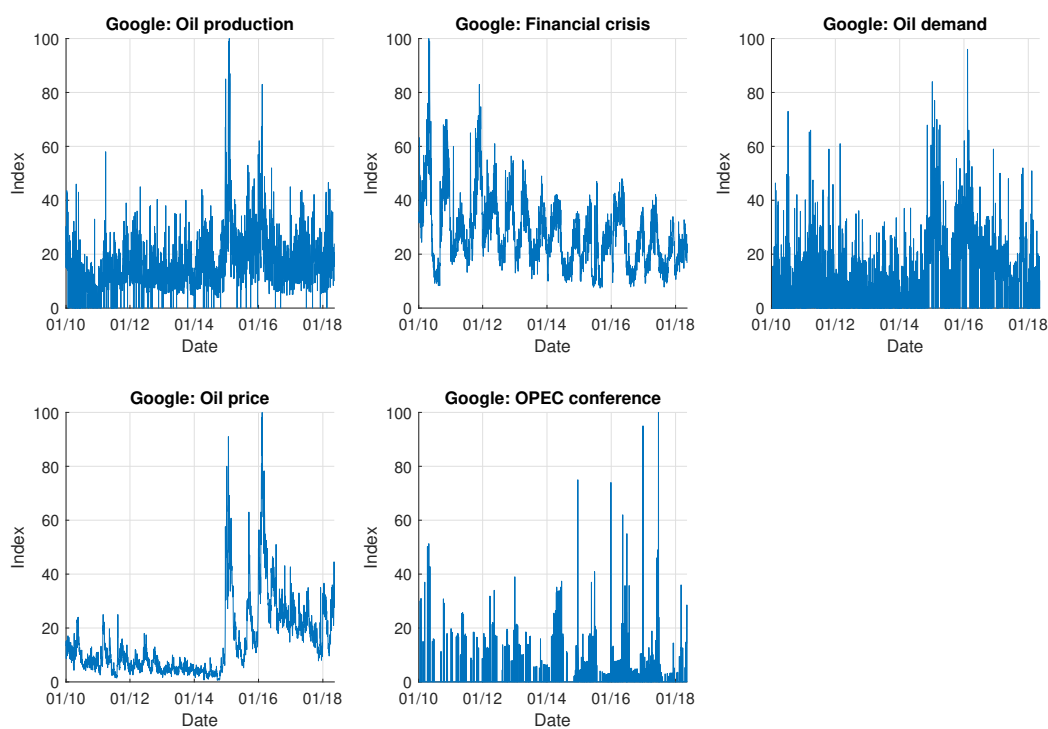


Figure B.4: Google search volume indices on daily resolution from January 5, 2010 to May 11, 2018, obtained from Google.